

Locality from Zeros

1 Background and motivation

Scattering amplitudes are the basic functions from which particle-scattering probabilities are computed. In standard quantum field theory, their structure is built from locality and unitarity: interactions occur through allowed local channels, and residues on physical poles factorize into lower-point amplitudes.

Here we ask whether these principles can instead emerge from hidden zeros: special kinematic loci where the amplitude vanishes. This is surprising because the zeros are a new property with no standard physical explanation yet (<https://arxiv.org/abs/2312.16282>), but they appear strong enough to remove non-local terms and fix the physical amplitude.

1.1 The theory: $\text{Tr } \phi^3$

We work in planar $\text{Tr } \phi^3$ theory, a simple scalar theory with a matrix-valued field ϕ and cubic interaction

$$\mathcal{L}_{\text{int}} = \frac{g}{3} \text{Tr}(\phi^3).$$

The trace gives a cyclic ordering of the external particles. At tree level, its Feynman diagrams are in one-to-one correspondence with triangulations of an n -gon: each internal chord gives a propagator $1/X_{ij}$.

This theory is useful because locality and factorization are completely transparent. Locality (physically=special relativity: interactions occur at points in spacetime) means that only non-crossing chords appear together, while unitarity (physically=quantum mechanics: probabilities add up to 1) means that residues on poles factorize into lower-point amplitudes. Thus the physics becomes a concrete algebraic-combinatorial question about which rational functions on the space of X -variables are allowed.

2 The basic dictionary

Fix n labelled points on a circle:

$$1, 2, \dots, n. \tag{1}$$

A variable X_{ij} is attached to the chord connecting i and j . We take $X_{ij} = X_{ji}$. Boundary edges are not counted as poles, so the interesting variables are the diagonal variables X_{ij} , with i, j not adjacent.

We also introduce the elementary finite-difference variables

$$c_{ij} = X_{i,j} + X_{i+1,j+1} - X_{i,j+1} - X_{i+1,j}, \quad (2)$$

with cyclic labels understood. These c_{ij} 's are the elementary plaquettes of the kinematic mesh. The hidden zeros are most naturally defined in terms of the c_{ij} 's.

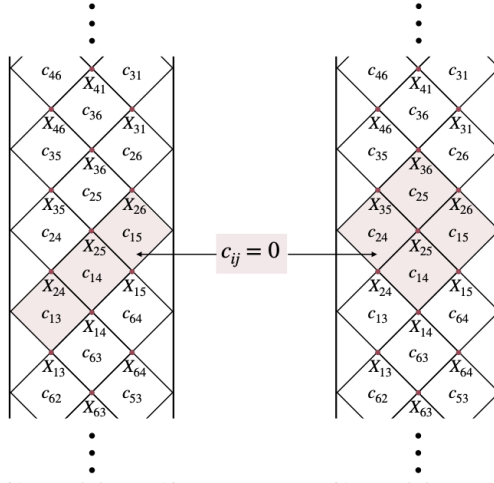


Figure 1: Kinematic mesh and two different zero conditions

2.1 Ordinary zeros

An ordinary zero Z_i is the locus on which all c 's in the i -th row of the mesh vanish, with the second index separated from i by at least two sites:

$$Z_i : \quad c_{i,j} = 0, \quad j = i + 2, i + 3, \dots, i + n - 2. \quad (3)$$

All labels are understood cyclically.

For example, at six points,

$$Z_1 : \quad c_{13} = c_{14} = c_{15} = 0. \quad (4)$$

Similarly,

$$Z_2 : \quad c_{24} = c_{25} = c_{26} = 0, \quad (5)$$

and so on.

2.2 Fat zeros

A fat zero Z_I , where I is a consecutive block of labels,

$$I = [a, b] = \{a, a + 1, \dots, b\}, \quad (6)$$

is defined by setting to zero a rectangular block of c 's:

$$Z_{[a,b]} : \quad c_{i,j} = 0, \quad i \in [a, b], \quad j \in [b + 2, a - 2]. \quad (7)$$

Again, all labels are cyclically ordered. The condition

$$j \in [b + 2, a - 2] \quad (8)$$

means that j lies outside the block I , and is not adjacent to either endpoint of I .

For example, at six points,

$$Z_{12} : \quad c_{14} = c_{15} = c_{24} = c_{25} = 0. \quad (9)$$

This is not the same thing as imposing both Z_1 and Z_2 . Indeed,

$$Z_1 : \quad c_{13} = c_{14} = c_{15} = 0, \quad (10)$$

while

$$Z_2 : \quad c_{24} = c_{25} = c_{26} = 0. \quad (11)$$

The fat zero Z_{12} instead keeps only the rectangular block with

$$i \in \{1, 2\}, \quad j \in \{4, 5\}. \quad (12)$$

Similarly, at six points,

$$Z_{23} : \quad c_{25} = c_{26} = c_{35} = c_{36} = 0, \quad (13)$$

and

$$Z_{34} : \quad c_{36} = c_{31} = c_{46} = c_{41} = 0. \quad (14)$$

For a larger example, still at six points,

$$Z_{123} : \quad c_{15} = c_{25} = c_{35} = 0. \quad (15)$$

Here the block is

$$i \in \{1, 2, 3\}, \quad j = 5. \quad (16)$$

Note this is equivalent to Z_5 .

2.3 Zeros and amplitudes

A triangulation of the n -gon is a maximal collection of mutually non-crossing chords. A triangulation has $n - 3$ chords. To such a triangulation T , we associate the rational term

$$\frac{1}{\prod_{X \in T} X}. \quad (17)$$

The planar cubic amplitude is the sum over all triangulations:

$$A_n = \sum_{\text{triangulations } T} \frac{1}{\prod_{X \in T} X}. \quad (18)$$

For example

$$A_4 = \frac{1}{X_{13}} + \frac{1}{X_{24}} \quad (19)$$

$$A_5 = \frac{1}{X_{13}X_{35}} + \text{cyclic} \quad (20)$$

A general ansatz B_n is allowed to be larger:

$$B_n = \sum_{S, |S|=n-3} \frac{c_S}{\prod_{X \in S} X}, \quad (21)$$

where S may contain crossing chords. If the chords in S are mutually non-crossing, the term is **local**. If S contains at least one crossing pair, the term is **non-local**.

The goal is to show that imposing a suitable collection of ordinary and fat zeros forces all non-local coefficients to vanish. After that, the remaining local ansatz is fixed by its residues and equals the physical amplitude.

$$\boxed{\text{ordinary and fat zeros} \implies \text{locality} \implies B_n = A_n.} \quad (22)$$

3 A five-point example

At five points, the diagonals are

$$X_{13}, \quad X_{24}, \quad X_{35}, \quad X_{14}, \quad X_{25}. \quad (23)$$

It is convenient to write them cyclically as

$$p_1 = X_{13}, \quad p_2 = X_{24}, \quad p_3 = X_{35}, \quad p_4 = X_{14}, \quad p_5 = X_{25}. \quad (24)$$

A local five-point term is a pair of non-crossing chords. The five local terms are

$$\frac{1}{p_1 p_3}, \quad \frac{1}{p_2 p_4}, \quad \frac{1}{p_3 p_5}, \quad \frac{1}{p_4 p_1}, \quad \frac{1}{p_5 p_2}. \quad (25)$$

Equivalently,

$$A_5 = \sum_{i=1}^5 \frac{1}{p_i p_{i+2}}, \quad (26)$$

where indices are cyclic modulo 5.

The non-local two-pole terms are the crossing pairs:

$$\frac{1}{p_1 p_2}, \quad \frac{1}{p_2 p_3}, \quad \frac{1}{p_3 p_4}, \quad \frac{1}{p_4 p_5}, \quad \frac{1}{p_5 p_1}. \quad (27)$$

Thus a general five-point ansatz with no higher-order poles may be written as

$$B_5 = \sum_{i=1}^5 a_i \frac{1}{p_i p_{i+2}} + \sum_{i=1}^5 b_i \frac{1}{p_i p_{i+1}}. \quad (28)$$

The coefficients a_i multiply local terms, while the coefficients b_i multiply non-local terms.

We now impose the hidden zero Z_1 . Recall that the elementary mesh variables are

$$c_{ij} = X_{i,j} + X_{i+1,j+1} - X_{i,j+1} - X_{i+1,j}. \quad (29)$$

At five points, the zero Z_1 is

$$Z_1 : \quad c_{13} = c_{14} = 0. \quad (30)$$

In terms of the p_i , these two equations are

$$c_{13} = X_{13} + X_{24} - X_{14} - X_{23} = p_1 + p_2 - p_4 = 0, \quad (31)$$

because X_{23} is a boundary edge, and

$$c_{14} = X_{14} + X_{25} - X_{15} - X_{24} = p_4 + p_5 - p_2 = 0, \quad (32)$$

because X_{15} is also a boundary edge. Therefore Z_1 imposes

$$p_4 = p_1 + p_2, \quad p_5 = -p_1. \quad (33)$$

Substituting this into the ansatz gives

$$\begin{aligned} B_5|_{Z_1} = & \frac{a_1}{p_1 p_3} + \frac{a_2}{p_2(p_1 + p_2)} - \frac{a_3}{p_1 p_3} + \frac{a_4}{p_1(p_1 + p_2)} - \frac{a_5}{p_1 p_2} \\ & + \frac{b_1}{p_1 p_2} + \frac{b_2}{p_2 p_3} + \frac{b_3}{p_3(p_1 + p_2)} - \frac{b_4}{p_1(p_1 + p_2)} - \frac{b_5}{p_1^2}. \end{aligned}$$

We require

$$B_5|_{Z_1} = 0 \quad (34)$$

as an identity in the remaining independent variables p_1, p_2, p_3 .

Putting everything over the common denominator

$$p_1^2 p_2 p_3 (p_1 + p_2),$$

the numerator is

$$\begin{aligned} & b_2 p_1^3 + (a_1 - a_3 + b_2 + b_3) p_1^2 p_2 + (a_1 - a_3) p_1 p_2^2 \\ & + (a_2 - a_5 + b_1) p_1^2 p_3 + (a_4 - a_5 + b_1 - b_4 - b_5) p_1 p_2 p_3 - b_5 p_2^2 p_3. \end{aligned}$$

Since this polynomial must vanish identically, every coefficient must vanish:

$$b_2 = 0,$$

$$a_1 - a_3 + b_2 + b_3 = 0,$$

$$a_1 - a_3 = 0,$$

$$a_2 - a_5 + b_1 = 0,$$

$$a_4 - a_5 + b_1 - b_4 - b_5 = 0,$$

$$b_5 = 0.$$

Equivalently,

$$\boxed{a_1 = a_3, \quad b_2 = b_3 = b_5 = 0, \quad b_1 = a_5 - a_2, \quad b_4 = a_4 - a_2.} \quad (35)$$

Thus a single zero Z_1 already eliminates three non-local coefficients:

$$b_2 = b_3 = b_5 = 0. \quad (36)$$

It also relates some local coefficients.

Repeating the step for the remaining zeros

$$B_5|_{Z_i} = 0, \quad i = 1, \dots, 5, \quad (37)$$

we find

$$a_1 = a_2 = a_3 = a_4 = a_5 \equiv a. \quad (38)$$

and

$$b_1 = b_2 = b_3 = b_4 = b_5 = 0. \quad (39)$$

Therefore the ansatz reduces to

$$B_5 = a \sum_{i=1}^5 \frac{1}{p_i p_{i+2}}. \quad (40)$$

Hence

$$B_5 = A_5 \quad (41)$$

This five-point example shows the two main effects of the zeros:

1. zeros eliminate crossing denominators, hence enforce locality;

(42)

2. after locality, zeros fix the amplitude, hence enforce the correct residues.

(43)

Statement 2 (assuming zeros and locality) was proven in <https://arxiv.org/abs/2406.04234> . The goal is to now prove statement 1.

4 Clean zeros, strict zeros, and active zeros

Let us make some useful definitions.

Let M be a set of poles on which we take residues. For example,

$$M = \{X_{16}, X_{27}, X_{37}\}. \quad (44)$$

Let Z be a zero. Algebraically, Z is defined by one or more linear equations:

$$L_\alpha = 0. \quad (45)$$

Definition 1 (Clean or strict zero). *A zero Z is clean, or strict, with respect to M , if after imposing*

$$X = 0 \quad \text{for all } X \in M, \quad (46)$$

the equations defining Z remain nontrivial and do not force any additional active pole $Y \notin M$ to vanish.

Equivalently, imposing the zero commutes with taking the residue:

$$\text{Res}_M(B|_Z) = (\text{Res}_M B)|_{Z,M}. \quad (47)$$

This is the useful property. If Z is clean with respect to M , then it descends directly to the coefficient of the residue. One can use this to form some inductive argument.

A zero may fail to be clean because it forces an extra pole Y to vanish. Then the residue is polluted by terms containing Y . However, if all terms containing Y have already been eliminated, the same zero becomes usable.

For example, if we have a zero condition $a + b = 0$, and we take a residue at $a = 0$, all terms containing b will now pollute the residue.

Definition 2 (Active zero). *A zero is called active for a given coefficient if it is clean after removing all terms that have already been shown to vanish.*

Thus cleanness is dynamic. A zero may not be clean at the beginning of the argument, but may become clean after other non-local terms have been removed.

5 Seven-point residue on X_{16}

We now illustrate the residue strategy at seven points. Assume again that no higher-order poles are present.

Take the parent residue

$$X_{16} = 0. \quad (48)$$

The only chords crossing X_{16} are

$$X_{27}, \quad X_{37}, \quad X_{47}, \quad X_{57}. \quad (49)$$

Thus every non-locality relative to the parent pole involves label 7.

A seven-point cubic term has four propagators. After taking the residue on X_{16} , three propagators remain. We organize the residue by pulling out explicitly all poles that cross X_{16} :

$$\text{Res}_{X_{16}=0} B_7 = B_6^{(3)}(1, 2, 3, 4, 5, 6) + \sum_{i=2}^5 \frac{C_i^{(2)}(1, 2, 3, 4, 5, 6)}{X_{i7}} + \sum_{2 \leq i < j \leq 5} \frac{C_{ij}^{(1)}(1, 2, 3, 4, 5, 6)}{X_{i7} X_{j7}} + \sum_{2 \leq i < j < k \leq 5} \frac{C_{ijk}^{(0)}(1, 2, 3, 4, 5, 6)}{X_{i7} X_{j7} X_{k7}}. \quad (50)$$

The notation is chosen so that $C^{(k)}$ has k remaining denominator factors. Thus:

$$C_i^{(2)} \text{ is a six-point two-pole function,} \quad (51)$$

$$C_{ij}^{(1)} \text{ is a six-point one-pole function,} \quad (52)$$

and

$$C_{ijk}^{(0)} \text{ is a constant.} \quad (53)$$

Crucially, all these functions depend only on the lower labels

$$1, 2, 3, 4, 5, 6. \quad (54)$$

This is again useful because it can enable inductive reasoning. The explicit factors X_{i7} encode the non-locality relative to the parent pole X_{16} .

Therefore, to prove locality near this parent pole, we need to show

$$C_i^{(2)} = 0, \quad C_{ij}^{(1)} = 0, \quad C_{ijk}^{(0)} = 0. \quad (55)$$

This removes every term containing a pole crossing X_{16} .

To prove full locality, this procedure must be repeated for every possible parent pole.

The question is then how to prove that all C functions are eliminated by zero conditions.

6 Strict zeros for the X_{16} residue

Let

$$\mathcal{I}_{16} = \{2, 3, 4, 5\}. \quad (56)$$

These labels correspond to the crossing poles

$$X_{27}, \quad X_{37}, \quad X_{47}, \quad X_{57}. \quad (57)$$

Consider a term whose explicit prefactor is

$$X_{16} \prod_{i \in S} X_{i7}, \quad S \subset \{2, 3, 4, 5\}. \quad (58)$$

The missing crossing poles are

$$\bar{S} = \{2, 3, 4, 5\} \setminus S. \quad (59)$$

The basic mesh rule is:

$$X_{a7} \text{ missing} \implies Z_{a-1}, Z_a \text{ are strict candidates.} \quad (60)$$

Thus

$$\mathcal{Z}_{\text{strict}}^{(1)}(S) = \bigcup_{a \in \bar{S}} \{Z_{a-1}, Z_a\}. \quad (61)$$

Fat zeros are included similarly. If the strict one-zero set contains a consecutive block

$$Z_a, Z_{a+1}, \dots, Z_b, \quad (62)$$

then the fat zero

$$Z_{a+1 \dots b} \quad (63)$$

is a natural strict candidate. However, the final test is always algebraic: reduce the equations defining the fat zero modulo the residue poles, and check that no additional active pole is forced to vanish.

For the $C^{(0)}$ terms, we get:

term	missing pole	strict one-zeros
$C_{234}^{(0)}$	X_{57}	Z_4, Z_5
$C_{235}^{(0)}$	X_{47}	Z_3, Z_4
$C_{245}^{(0)}$	X_{37}	Z_2, Z_3
$C_{345}^{(0)}$	X_{27}	Z_1, Z_2

(64)

Since $C^{(0)}$ has no remaining denominator factors, any nontrivial strict zero forces it to vanish. Therefore

$$C_{234}^{(0)} = C_{235}^{(0)} = C_{245}^{(0)} = C_{345}^{(0)} = 0. \quad (65)$$

For the $C^{(1)}$ terms:

coefficient	strict one-zeros
$C_{23}^{(1)}$	Z_3, Z_4, Z_5
$C_{24}^{(1)}$	Z_2, Z_3, Z_4, Z_5
$C_{25}^{(1)}$	Z_2, Z_3, Z_4
$C_{34}^{(1)}$	Z_1, Z_2, Z_4, Z_5
$C_{35}^{(1)}$	Z_1, Z_2, Z_3, Z_4
$C_{45}^{(1)}$	Z_1, Z_2, Z_3

(66)

Some $C^{(1)}$ functions obey more strict conditions than others. This is useful: one should eliminate the most constrained coefficients first. After they are removed, further zeros may become active for the remaining coefficients.

For the $C^{(2)}$ terms:

coefficient	strict one-zeros
$C_2^{(2)}$	Z_2, Z_3, Z_4, Z_5
$C_3^{(2)}$	Z_1, Z_2, Z_3, Z_4, Z_5
$C_4^{(2)}$	Z_1, Z_2, Z_3, Z_4, Z_5
$C_5^{(2)}$	Z_1, Z_2, Z_3, Z_4

(67)

Thus each $C_i^{(2)}$ has at least four strict one-zeros, and often several fat-zero constraints. If the six-point base statement says that a two-pole six-point function cannot satisfy four suitable zeros, then all $C_i^{(2)}$ vanish.

7 Strategy 1: shifts

One possible strategy is to use shifts adapted to the zeros.

Suppose a zero equation has the schematic form

$$X_{ij} = X_{ab} + X_{cd}. \quad (68)$$

A shift preserving this zero is, for example,

$$X_{ab} \mapsto X_{ab} + t, \quad X_{cd} \mapsto X_{cd} - t, \quad t \rightarrow \infty. \quad (69)$$

The zero condition remains true along the shift.

Now apply such a shift to the ansatz. Many terms vanish as $t \rightarrow \infty$. The terms that survive must cancel, because the full ansatz vanishes on the zero.

This is especially clean for the $C^{(0)}$ terms. We just need a shift that avoids all X in the prefactor. Such a shift will eliminate that term. This is doable, because of the universal form of the prefactor.

For the higher k C , this becomes more difficult.

8 Strategy 2: strict zeros and induction

The second strategy is to use strict zeros and induction.

The residue expansion gives lower-point functions:

$$C_i^{(2)}(1, \dots, 6), \quad C_{ij}^{(1)}(1, \dots, 6), \quad C_{ijk}^{(0)}(1, \dots, 6). \quad (70)$$

The hope is that each such coefficient inherits enough strict or active zeros to be eliminated by a lower-point statement.

A useful induction target is:

$$\boxed{C_m^{(k)} \text{ with } k \text{ denominator factors cannot satisfy enough active zeros unless it vanishes.}} \quad (71)$$

For example, one needs base statements such as:

$$C_6^{(0)} \text{ cannot satisfy two suitable zeros,} \quad (72)$$

$$C_6^{(1)} \text{ cannot satisfy three suitable zeros,} \quad (73)$$

$$C_6^{(2)} \text{ cannot satisfy four suitable zeros.} \quad (74)$$

The last statement is especially important, because higher-point induction often descends to a six-point two-pole function $C_6^{(2)}$.

The active-zero idea is important for making induction plausible. Suppose a zero is not strict for a coefficient C because it also forces an additional pole Y to vanish. Then C is polluted by terms containing Y . But if those terms have already been eliminated, then the same zero becomes active for C . Thus the proof should proceed triangularly:

$$C^{(0)} \longrightarrow C^{(1)} \longrightarrow C^{(2)} \longrightarrow \dots \quad (75)$$

In the seven-point X_{16} residue, this means:

$$C^{(0)} \text{ terms vanish first;} \quad (76)$$

then zeros polluted only by $C^{(0)}$ terms become active for $C^{(1)}$; then zeros polluted only by $C^{(1)}$ terms become active for $C^{(2)}$.

This is weaker than demanding that every zero be strict from the start, and is therefore more likely to close.